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A comparative study of capacity market demand curve designs considering risk-averse market participants

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***Abstract*—While capacity remuneration mechanisms are emerging around the globe, a variety of capacity demand curves is being used. Poorly tuned capacity demand curves can, however, lead to expensive over- or underprocurement of capacity. This work provides a methodology to directly compare different capacity demand curve design methodologies, accurately reflecting the net cost of new entry in the definition of the demand curves. To this end, a long-run equilibrium with energy-only and capacity markets with risk-averse market participants is considered. We find that sloped capacity demand curves, setting the net cost of new entry at the reliability target, and inelastic capacity demands, that are set to the reliability target, perform ideally assuming perfect foresight. Considering erroneous assumptions on the value-of-lost-load and risk-aversion, the results obtained using these sloped capacity demand curves are more sensitive to uncertainties than those using inelastic capacity targets. In contrast, conservatively tuned marginal reliability impact-based curves perform suboptimal assuming perfect foresight, but are less sensitive to erroneous parameterization.**

***Index Terms*—Capacity Markets, Risk aversion, Capacity demand curve, Net cost of new entry**

I. INTRODUCTION

IN theory, electricity spot markets provide social optimal levels of capacity assuming rational market participants acting on perfect and complete markets without price caps [1]. Taking involuntarily load shedding into account, optimal levels of capacity are secured if price caps are set at the Value-Of-Lost-Load (VOLL) [2]. In reality, factors such as price caps, set below or at erroneous estimates of the VOLL [3], and risk aversion lead to a suboptimal market outcomes [4]. Price caps have been introduced to mitigate price risks for consumers, but may weaken the investment incentives and could, therefore, cause underinvestments [5]. Similarly as capital markets price risk, capital costs for risk-averse investments in electricity generation capacity exceed that of a benevolent planner [6], leading to a suboptimal investment equilibrium from a social perspective.

To avoid underinvestments in capacity and ensure generation system adequacy, Capacity Remuneration Mechanisms (CRMs) are implemented [7]. CRMs should ensure that the reliability standard, i.e. targets in terms of Expected Energy Not Served (EENS) or Loss Of Load Expectation (LOLE), are satisfied. CRMs can be categorized as price-based and quantity-based mechanisms. While in price-based mechanisms, e.g. capacity payments, usually a fixed compensation per MW of generation capacity is provided, the compensation in quantity-based mechanisms, such as strategic reserves or centralized capacity markets, is usually determined through a market-based approach. In the latter approach, capacity demand curves express the willingness to pay for capacity [8]. As such, the Capacity Demand Curve (CDC) in a capacity market is intimately linked with the reliability standard, which is based on the VOLL and the (net) Cost Of New Entry (CONE) as, e.g., in Europe [9, Article 25]. The net Cost Of New Entry (net-CONE) is the difference between the CONE and the net revenues from energy and ancillary services markets [10] and, therefore, a system-dependent parameter in the definition of the CDCs.

Although numerous modeling approaches focusing on electricity market design and capacity remuneration mechanisms exist [11], just a handful of approaches consider risk-averse investors, which can be classified as stochastic equilibrium models, system dynamics models, or agent-based models. In the first group, Ehrenmann et al. [12] solved a stochastic equilibrium model formulated as a mixed complementarity problem using a fixed capacity target concluding that capacity markets are better suited to guarantee the generation system adequacy while shifting the generation mix to less capital-intensive technologies. Höschle et al. [13] modeled a downward sloping CDC confirming previous findings also for markets with high price caps. Mimicking an inelastic capacity demand by call options, Mays et al. [14] describe an asymmetric effect of capacity markets on the risk profile of technologies. Mays et al. confirm the shift towards technologies with lower fixed costs and higher operating costs, concluding that the current market structures may be ill-suited to financing low-carbon resources. Similarly, using a fixed capacity target, Kaminski et al. [15] find that capacity markets are needed even if energy demand becomes elastic. In addition, Shu and Mays explore alternative resource adequacy contract designs and differentiate these from approaches considering capacity mechanisms [16]. In the second group, using system dynamics models, comparisons of an Energy-Only Market (EOM) and an EOM complemented with a capacity market with fixed capacity demand showed that CRMs are more resilient w.r.t. risk aversion in terms of cost and reliability [17], [18], [19]. Lastly, using an agent-based approach to study bilateral contracts on an EOM with an inelastic CRM demand, Tao et al. [20] observe more stable levels of installed capacity with a CRM. Extending to long-term uncertainties in a multi-country case study of the European electricity market, Fraunholz et al. [21] confirm that risk-averse investors exhibit reduced investments in capacity.

While poorly tuned CDCs can lead to expensive over- or underprocurement of capacity [15], no contribution exists that assesses the impact of different shapes of capacity demand curves on system performance while considering risk aversion and the net-CONE. Indeed, most scientific work focuses on inelastic capacity demand curves, whereas in real-life CRMs a multitude of CDC designs have been adopted (see Section II). In this work, we focus on a coherent comparison of different capacity demand curve definitions and the performance of these different CDCs in terms of system costs. This framework allows assessing the impact of regulatory/political design choices.

As CDC definitions are typically based on the reliability standard, defining price-quantity pairs connected to the CONE or net-CONE, these definitions do not establish a direct connection between capacity compensation and social welfare. Therefore, we additionally propose a novel, strictly economically motivated CDC that is based on marginal social benefit and compare it with well-established CDCs. The contribution of this work is threefold:

1) We propose a novel methodology to reflect the net-CONE in the definition of CDCs in Generation Expansion Planning (GEP) problems, which allows a coherent comparison of different CDCs;
2) We present a novel method for defining CDCs reflecting the marginal social benefit of capacity;
3) We provide a quantitative comparison of selected capacity demand curves definitions using a stylized generation expansion planning problem with risk-averse actors assuming a perfect and imperfect information base. Capacity demand curves that contain the target capacity of the reliability target at the price of the net-CONE perform ideally with perfect foresight. With an imperfect information base, results show that sloped CDCs are more sensitive to uncertainties leading to higher system costs with low VOLLs than inelastic CDCs.

This paper is structured as follows. Section II introduces the selection of CDCs considered in this work. Section III describes the modeling approach that is used in the case study in Section IV. Section V concludes.

II. Capacity remuneration mechanisms and demand curves

Capacity markets remunerate capacity by a per-MW payment in order "to eliminate residual adequacy concerns" [9]. Figure 1 gives a qualitative overview of the system costs depending on the amount of installed capacity, illustrating the capacity level an optimal CDC ideally promotes. Considering a deficit in capacity (Fig. 1, left), increased costs by involuntarily load shedding occur. Due to the typical shapes of load-duration curves, the cost savings by curtailing less energy demand decrease with higher levels of installed capacity. Assuming an excess of capacity (Fig. 1, middle), the system costs increase, driven by additional costs for installing and operating capacity minus cost savings by reduced ENS. However, the increase in costs of a capacity deficit is typically higher than that of excess capacity assuming the same deviation from the optimal capacity level. If the installed capacity exceeds the

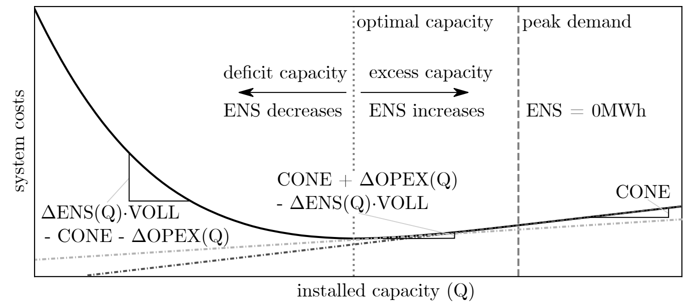

Figure 1: System costs (i.e., CAPEX, OPEX and costs of lost load (product of Energy Not Served (ENS) and the Value Of Lost Load (VOLL)) as a function of installed capacity Q.

peak demand (Fig. 1, right), the costs for installing excess capacity (CONE) is the only cost-driver.

To determine the optimal adequacy level and, hence, the optimal level of installed capacity in the definition of CDCs, in Europe, Reliability Standards (RS) are used, taking the VOLL and the CONE into account [9], [22]. The VOLL is synthesized using the Expected Energy Not Served (EENS) weighted VOLL per consumer. The CONE contains a variable part (connecting to the OPEX) and a fixed part (annualized capital costs, annual fixed costs) that takes the de-rating capacity factor into account. Both VOLL and CONE are used to express the LOLE per technology, of which the smallest (available) LOLE constitutes the reliability standard. Assuming constant VOLL, the RS calculation simplifies to:

$$\mathrm{LOLE}_{\mathrm{RS}} = \frac{\mathrm{CONE}}{\mathrm{VOLL} - \mathrm{OPEX}}. \tag{1}$$

This approach is motivated by the fact that under idealized circumstances, a social planner would construct a system that is characterized by the same LOLE [2]. Hence, we refer to the reliability standard as the social optimal capacity target.

Although the resulting capacity demand could be enforced directly in a CRM under the form of a inelastic CDC, we observe a transition from inelastic to elastic demand curves in real-life power systems [10]. Sloped demand curves were introduced as they promise to be more stable in terms of prices [23], but also to complexify the anticipation of market outcomes to hamper strategic power abuse [24]. The parameterization of a CDC is a very complex task, requiring the correct estimation of system-specific parameters, such as VOLLs, net-CONE (which is connected to revenues from energy and ancillary markets), interconnection capacity, forced downtimes, installed capacity, capacity factors, etc.

In this work, six representative CDCs definitions are discussed and quantitatively compared aiming to reflect the wide spectrum of real-life CDCs. These CDCs are: an inelastic capacity demand, a hypothetical curve that is based on Marginal Social Welfare Impact (MSWI) (Section III-E), a CDC proposed by Zhao et al. [25] (referred to as *Marginal-Reliability-Impact (MRI)* curve), an ISO-NE-inspired CDC (referred to as *Scaled Marginal-Reliability-Impact (SMRI)* curve), a PJM-inspired CDC (referred to as *elastic* curve), and a demand curve motivated by the sloped CDC used in Belgium (referred to as *semi-elastic* curve). To ensure a fair

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and coherent comparison of the CDCs, the CDC definitions used in this work are based on the principles of capacity demand curves applied in reality, while the level of detail is kept stylized, constituting significant simplifications from their real-world counterparts (Section III-C).

III. Methodology

To perform a coherent comparison of the considered CDCs, a generation expansion planning problem formulated as an equilibrium problem including risk-averse market participants is developed (Section III-A). The solution approach is explained in Section III-B. The parameterization of the CDCs is presented in Section III-C. As the net-CONE is calculated taking the revenues of the EOM into account, it is dependent on the resulting electricity system. To reflect this dependency, we present an iterative approach to accurately capture the net-CONE in the definition of CDCs in the long-term equilibrium (Section III-D). Lastly, a novel method for defining CDCs is introduced reflecting the marginal social benefit of capacity (Section III-E).

A. Generation expansion planning problem

In a stylized fashion, the risk-averse generation expansion planning problem can be mathematically modeled as an annualized equilibrium problem with competitive participants considering no other market distortions than price caps and risk aversion. Here, the consumer side is simplified into an aggregated consumer having a single VOLL. The producer side is represented by Generation Companies (GenCos) that invest each in a single technology. In the formulation below, $\mathrm{P_s}$ indicates the probability of scenario, s, and $\mathrm{W_t}$ indicates the length of a period, t. k is an index for generation technologies. $\mathbb{E}(\cdot)$ calculates the expected outcome as $\mathbb{E}(X) = \sum_{\mathrm{s}} \mathrm{P_s} X_{\mathrm{s}}$. Dual variables are indicated between parentheses.

1) Generation Companies

Each risk-averse GenCo gc maximizes the γ-weighted expected surplus (Eq. (2.1.8)) and a $(1-\gamma)$-weighted Conditional Value-At-Risk (CVAR, Eq. (2.1.9)). The surplus is defined as the difference between the revenues ($R_{\mathrm{gc,s}}$, second line Eq. (2.1.8)) from the capacity and the energy market and costs ($C_{\mathrm{gc,s}}$, third line Eq. (2.1.8)), of producing energy and installing capacity. The energy and capacity prices are denoted by $\lambda^{\mathrm{e}}_{\mathrm{t,s}}$ and λ^{cm}. The CVAR (Eq. (2.1.9)) is modeled following Rockafellar [26] describing the expectation of all scenarios that belong to the worst-tail. The length of the tail, i.e., the degree of risk aversion, is defined by the parameter $\beta \in [0,1]$. A low value of β indicates a short tail, i.e., risk-averse GenCos, while the CVAR is equal to the expected surplus with $\beta = 1$.

$$\max_{\substack{\mathrm{e_{gc,k,t,s}},\mathrm{cm_{gc,k}},y_{\mathrm{gc,k}},\\ \alpha_{\mathrm{gc}},u_{\mathrm{gc,s}}}} \gamma \mathbb{E}(S_{\mathrm{gc}}) + (1-\gamma)\mathrm{CVAR_{gc}} \qquad (2.1.1)$$

s.t.

$$0 \le \mathrm{e_{gc,k,t,s}} \le y_{\mathrm{gc,k}} \cdot \mathrm{AV^{e}_{gc,k,t,s}} \quad (\overline{\delta}^{\mathrm{e}}_{\mathrm{gc,k,t,s}}) \quad \forall \mathrm{t,s} \qquad (2.1.2)$$

$$0 \le \mathrm{cm_{gc,k}} \le y_{\mathrm{gc,k}} \cdot \mathrm{AV^{cm}_{gc,k}} \quad (\overline{\delta}^{\mathrm{cm}}_{\mathrm{gc,k}}) \qquad (2.1.3)$$

$$0 \le y_{\mathrm{gc,k}} \qquad (2.1.4)$$

$$\alpha_{\mathrm{gc}} \in \mathbb{R} \qquad (2.1.5)$$

$$0 \le u_{\mathrm{gc,s}} \quad \forall \mathrm{s} \qquad (2.1.6)$$

$$\alpha_{\mathrm{gc}} - S_{\mathrm{gc,s}} \le u_{\mathrm{gc,s}} \quad (\delta_{\mathrm{gc,s}}) \quad \forall \mathrm{s} \qquad (2.1.7)$$

with

$$\begin{aligned} S_{\mathrm{gc,s}} &= R_{\mathrm{gc,s}} - C_{\mathrm{gc,s}} \qquad (2.1.8)\\ &= \mathrm{cm_{gc,k}} \cdot \lambda^{\mathrm{cm}} + \sum_{\mathrm{t}} \mathrm{e_{gc,k,t,s}} \cdot \mathrm{W_t} \cdot \lambda^{\mathrm{e}}_{\mathrm{t,s}} \\ &\quad - \sum_{\mathrm{t}} \mathrm{e_{gc,k,t,s}} \cdot \mathrm{W_t} \cdot \mathrm{OPEX_{k,t,s}} - y_{\mathrm{gc,k}} \cdot \mathrm{CAPEX_k} \end{aligned}$$

$$\mathrm{CVAR_{gc}} = \alpha_{\mathrm{gc}} - \frac{1}{\beta}\mathbb{E}(u_{\mathrm{gc}}). \qquad (2.1.9)$$

Constraint (2.1.2) limits the generation of electricity, $\mathrm{e_{gc,k,t,s}}$, to the available installed capacity, $y_{\mathrm{gc,k}}\mathrm{AV^{e}_{gc,k,t,s}}$. Similarly, Eq. (2.1.3) constrains offered capacity, $\mathrm{cm_{gc,k}}$, to the capacity market to the derated capacity ($y_{\mathrm{gc,k}}\mathrm{AV^{cm}_{gc,k}}$). Therefore, GenCos act as price-takers, submitting bids that reflect the marginal costs of generating electricity or providing generation capacity.

To describe the conditional value-at-risk linearly, the introduction of auxiliary variables (α and u_{s}) is necessary [26]: α reflects the value at risk, while u_{s} can be interpreted as the surplus difference w.r.t. the VAR for the scenarios belonging to the worst-case tail (cf. Eq. (2.1.7)).

2) Consumer

The consumer maximizes a weighted sum of its surplus and its CVAR. The surplus is defined as the difference between the utility associated with procuring energy and capacity (Eq. (2.2.7), first line) and the costs for capacity and electricity (Eq. (2.2.7), second line).

$$\max_{\substack{\mathrm{e_{co,t,s}},\mathrm{cm}_{\mathrm{co},c},\\ \alpha_{\mathrm{co}},u_{\mathrm{co,s}}}} \gamma \mathbb{E}(S_{\mathrm{co}}) + (1-\gamma)\mathrm{CVAR_{co}} \qquad (2.2.1)$$

s.t.

$$0 \le \mathrm{e_{co,t,s}} \le \overline{D^{\mathrm{e}}_{\mathrm{t,s}}} \quad (\overline{\mu}^{cap}_{\mathrm{t,s}}) \quad \forall \mathrm{t,s} \qquad (2.2.2)$$

$$0 \le \mathrm{cm}_{\mathrm{co},c} \le \overline{D^{\mathrm{cm}}_{c}} \quad (\overline{\delta}^{\mathrm{cm}}_{\mathrm{co},c}) \quad \forall c \qquad (2.2.3)$$

$$\alpha_{\mathrm{co}} \in \mathbb{R} \qquad (2.2.4)$$

$$0 \le u_{\mathrm{co,s}} \quad \forall \mathrm{s} \qquad (2.2.5)$$

$$\alpha_{\mathrm{co}} - S_{\mathrm{co,s}} \le u_{\mathrm{co,s}} \quad (\delta_{\mathrm{co,s}}) \quad \forall \mathrm{s} \qquad (2.2.6)$$

with

$$\begin{aligned} S_{\mathrm{co,s}} &= \sum_{\mathrm{t}} \mathrm{W_t} \cdot \mathrm{e_{co,t,s}} \cdot \overline{\lambda^{\mathrm{e}}} + \sum_{c} \mathrm{WTP}^{\mathrm{cm}}_{c} \cdot \mathrm{cm}_{\mathrm{co},c} \\ &\quad - \sum_{\mathrm{t}} \mathrm{W_t} \cdot \mathrm{e_{co,t,s}} \cdot \lambda^{\mathrm{e}}_{\mathrm{t,s}} - \sum_{c} \lambda^{\mathrm{cm}} \cdot \mathrm{cm}_{\mathrm{co},c} \qquad (2.2.7) \end{aligned}$$

$$\mathrm{CVAR_{co}} = \alpha_{\mathrm{co}} - \frac{1}{\beta}\mathbb{E}(u_{\mathrm{co}}) \qquad (2.2.8)$$

The utility associated with procuring capacity is formally the integration of the willingness-to-pay for capacity, which is defined by the CDC. Capacity demand curves are typically monotonously decreasing (see Fig. 2), which allows a

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piecewise-linear approximation. Therefore, the capacity demand curve is discretized in a finite number of sections, c. Each section takes a scalar willingness-to-pay ($\mathrm{WTP}_c^{\mathrm{cm}} \geq \mathrm{WTP}_{\mathrm{cm},c+1}$). The consumption of capacity of each section, $\mathrm{cm}_{\mathrm{co},c}$, is constrained by an upper bound indicating the section's boundary, $\overline{D_c^{\mathrm{cm}}}$ (Eq. (2.2.3)). The utility of consuming energy is limited by the VOLL or the price cap ($\overline{\lambda^{\mathrm{e}}}$), if the price cap is set below the VOLL. The consumed energy, $\mathrm{e}_{\mathrm{co,t,s}}$, is constrained by the inelastic energy demand, $\overline{D_{\mathrm{t,s}}^{\mathrm{e}}}$. Risk aversion is described analogously to Problem (2.1).

3) Market balances

An hourly energy market and a yearly centralized capacity market connect the GenCos' and consumer's problems:

$$(\mathrm{e}_{\mathrm{co,t,s}} - \sum_{\mathrm{k}} \mathrm{e}_{\mathrm{gc,k,t,s}}) \cdot \mathrm{W_t} = 0 \qquad \forall \mathrm{t,s} \qquad (2.3.1)$$

$$\sum_{c} \mathrm{cm}_{\mathrm{co},c} - \sum_{\mathrm{k}} \mathrm{cm}_{\mathrm{gc,k}} = 0. \qquad (2.3.2)$$

B. Solution strategy

Risky design equilibrium problems with incomplete risk-trading are non-linear and include non-convex, non-smooth objectives (CVAR), which often cause a multiplicity of equilibriums and are hard to solve [27], [28]. However, if all agents choose the same set of scenarios as worst-case scenarios, as we will assume in this paper, the risky design equilibrium coincides with a risky design game [29]. In such cases, the risky-design game can be modeled as an optimization problem using a risk-averse central planner, see, e.g., [30], [14].

In this work, we consider solely uncertainty in the electricity demand. Scenarios are constructed by scaling a given load-duration curve (multiplying by a given factor) resulting in the shape of the demand curve to remain the same. Therefore, both the consumer and the generation companies favor high-demand scenarios over low-demand scenarios, which justifies the assumption of a shared set of worst-case scenarios for the consumer and the GenCos. For the producer having lower energy demand entails having lower profit given an investment decision. For the consumer, lower energy demand leads to lower utility from consuming energy. Hence, in this work, the presented risky design equilibrium can be formulated as a risky design game. To recast the risky design game as a central planner problem, the surplus maximization problems of the consumer (Problem 2.2) and generation companies (Problem 2.1) are aggregated by summing the single participants' objectives and merging the constraints into a single problem. Revenue terms of the producers are equivalent to the costs of the consumer and therefore eliminate each other. Market balances (Eq. 2.3) are internalized and the associated dual variables can be interpreted at the market prices weighted by the probability adjusted for risk aversion ($\gamma \mathrm{P_s} + \delta_{\mathrm{gc,s}}$). The congruence of the risk-averse equilibrium problem with the risk-averse central planner problem can be shown by comparing the KKT conditions of the equilibrium problem with those of the central planner problem, not presented here for the sake of brevity. The resulting centralized optimization is linear and convex for all monotonic decreasing capacity demand curves. Therefore, the proposed model can facilitate the use of a high temporal resolution to precisely reflect times of scarcity and can be applied to a wide range of CDC definitions.

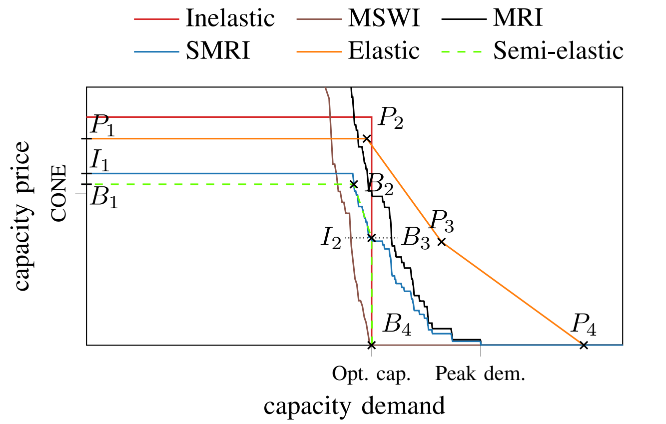

Figure 2: Illustration of parametrization of capacity demand curves assuming equilibrium net-CONE

C. Parametrization of existing capacity demand curves

This work focuses on the VOLL and the net-CONE in the definitions of the CDCs. Other details in the CDC definitions are simplified as much as possible: we assume the availability and the capacity derating factors in the capacity markets to be one and we refrain from considering ancillary markets and interconnections. The remaining parameters, which are CONE, VOLL, the net-CONE, and the associated reliability standard are enforced coherently for every considered CDC.

The CONE is assumed to be the annualized cost of capital. The reliability standard is not dependent on the choice of the CDC, but on the VOLL, CONE, and OPEX as introduced in Eq. (1). The net-CONE is defined as CONE minus Net Energy and Ancillary Services (E&AS) Revenue Offset averaged over the last one to three years, depending on the definition [31], [9]. Consequently, the net-CONE is a very system-specific value being impacted by the amount of capacity in the system. We propose an iterative procedure to calculate the net-CONE (Section III-D) to ensure that it correctly reflects the cost of new entry in the long-run equilibrium. To perform a coherent comparison, we consider a greenfield approach and we use identical CONE and net-CONE calculation approaches (Section III-D) for all CDCs.

We consider five CDCs (Fig. 2) from the literature and real-life capacity markets, which are parameterized as follows (the sixth CDC, the marginal social welfare-based CDC developed in this paper, is introduced in Section III-E):

1) The **inelastic** capacity target is set according to the reliability standard, hence, at the socially optimal capacity. A capacity price cap is considered at 1.5-times CONE.

2) The Marginal-Reliability-Impact (**MRI**) based curve is based on the loss of load expectation (LOLE), which equals the change in Expected Energy Not Served (EENS) w.r.t. the change in available capacity (Q) [4].

$$\mathrm{LOLE}(Q) = -\frac{\partial EENS(Q)}{\partial Q}. \qquad (4)$$

Zhao argues that the value of reliability for the corresponding system, hence, the capacity price, can be represented by the LOLE(Q) multiplied by the VOLL [25]:

$$\lambda^{\mathrm{cm}}_{\mathrm{zhao}} = -\frac{\partial EENS(Q)}{\partial Q} \cdot \mathrm{VOLL}. \tag{5}$$

3) The Scaled MRI (**SMRI**) curve constitutes the MRI curve multiplied by a factor such that the curve intersects with the net-CONE at the Net Installed Capacity Requirement (NICR), which is assumed to be the reliability standard, point I_2. Furthermore, the maximum capacity price is capped by $max(\mathrm{CONE}, 1.6 \cdot \text{net-CONE})$, cf. point I_1 in Fig. 2 [10], [23]. The SMRI is inspired by the CDC as used by the ISO-NE.
4) The **elastic** capacity demand curve is inspired by the principles of the CDC as used in the PJM market zone [32], [10]. In this work, the reliability requirement (RR) is set to the reliability standard. The curve is defined by a price cap at $max(\mathrm{CONE}, 1.5 \cdot \text{net-CONE})$ and three additional points (P_2, P_3, and P_4). The ordinates of these points are set to the price cap, $0.75 \cdot \text{net-CONE}$ and $0\,\frac{€}{\mathrm{MWh}}$. The abscissas of this curve are here defined according to the 2018-2022 delivery year settings as $0.998\mathrm{RS}$, $1.029\mathrm{RS}$, and $1.088\mathrm{RS}$.
5) The definition of the **semi-elastic** capacity demand curve is derived from "State aid SA.54915 (2019/N)" as utilized in Belgium [33] using three points (B_2, B_3, and B_4) and a price cap that is reflected by net-CONE multiplied by a factor X (here: 1.5). The reliability target is defined by the LOLE from the RS. Here, B_2 is defined as $(Q(\mathrm{X} \cdot \mathrm{LOLE_{RS}}), \mathrm{X} \cdot \text{net-CONE})$, B_3 as $(Q(\mathrm{LOLE_{RS}}), \text{net-CONE})$, and B_4 as $(Q(\mathrm{LOLE_{RS}}), 0.0)$. The net-CONE and the target reliability are multiplied with the factor X, leading to the semi-elastic curve intersecting with the SMRI curve at point B_2.

D. Net-CONE in the long-term equilibrium

Net-CONEs represent the missing money that generators have to secure in capacity markets, here defined as CONE corrected for the inframarginal rents from electricity generation [31]. Therefore, the capacity market price must equal the net-CONE in the long-term equilibrium. This connection can be formally proved via the KKT conditions of Problem (2), but has been left out for sake of brevity.

Capacity demand curves are often defined taking net-CONEs into account (cf. Sec. II). However, the capacity demand curve design impacts the amount of capacity in the system and, therefore, the missing money leads to a chicken-and-egg problem: the definition of the capacity demand curve impacts the amount of capacity and, hence, the revenues from the electricity market. The amount of capacity impacts the missing money and, therefore, the capacity demand curve. Risk aversion impacts the missing money in opposite ways. On the one hand, the degree of risk aversion increases the required risk premium. On the other hand, risk aversion possibly leads to capacity deficits, increasing scarcity rents. Hence, ex-ante calculations of the net-CONEs do not allow a coherent comparison as these are system-specific. This challenges the formal comparison of CDCs, as long-term net-CONEs of different curves are not known.

To enable a coherent and fair comparison of CDCs in the long-term equilibrium, in this work, a novel iterative approach to find the long-term missing money is proposed (Algorithm 1) and applied to align the net-CONE with the capacity market price. In Algorithm 1, the net-CONE is lowered by a decrement, ϵ, and compared with the price of the capacity market as long as the absolute error between capacity price and net-CONE is smaller than the decrement. Being an iterative approach, the choice of ϵ is a compromise between performance (large ϵ) and solution quality (small ϵ). Here, $\epsilon = 20\,\frac{€}{\mathrm{MW}}$ is chosen, which is 3 orders of magnitude smaller than typical prices on the capacity market.

Algorithm 1 Find equilibrium net-CONE

```
net-CONE ← CONE
λ^cm ← 0
while |net-CONE − λ^cm| ≥ ϵ do
    net-CONE ← net-CONE − ϵ
    λ^cm ← solve(Problem (2))
end while
```

E. Marginal social welfare impact curve (MSWI)

Capacity demand curve definitions typically follow reliability requirements defining price-quantity pairs connected to the CONE or net-CONE. To establish a direct connection between capacity compensation and social welfare, we propose a strictly economically motivated CDC definition. To this end, we use a centralized optimization problem that considers an energy-only market (Problem (6)) with a binding capacity constraint (Eq. (6.6)). The dual of this constraint, κ, can be interpreted as the marginal value of capacity for society.

$$\max SW = \sum_{\mathrm{t,s}} \mathrm{P_s W_t} \Bigg(\mathrm{VOLL} \cdot \mathrm{e_{co,t,s}} - \sum_{\mathrm{k}} \mathrm{e_{gc,k,t,s}} \cdot \mathrm{OPEX_{k,t,s}} \Bigg) - \sum_{\mathrm{k}} y_{\mathrm{gc,k}} \mathrm{CAPEX_k} \tag{6.1}$$

$s.t.:$

$$0 \le \mathrm{e_{gc,k,t,s}} \le y_{\mathrm{gc,k}} \mathrm{AV^e_{gc,k,t,s}} \qquad (\overline{\delta}^{\mathrm{e}}_{\mathrm{gc,k,t,s}}) \quad \forall \mathrm{k,t,s} \tag{6.2}$$

$$0 \le y_{\mathrm{gc,k}} \qquad (\overline{\delta}^{cap}_{\mathrm{gc,k}}) \quad \forall \mathrm{k} \tag{6.3}$$

$$0 \le \mathrm{e_{co,t,s}} \le \overline{D^{\mathrm{e}}_{\mathrm{t,s}}} \qquad (\overline{\mu}^{cap}_{\mathrm{t,s}}) \quad \forall \mathrm{t,s} \tag{6.4}$$

$$\mathrm{W_t} \cdot (\mathrm{e_{co,t,s}} - \sum_{\mathrm{k}} \mathrm{e_{gc,k,t,s}}) = 0 \qquad (\lambda^{\mathrm{e}}_{\mathrm{t,s}}) \quad \forall \mathrm{t,s} \tag{6.5}$$

$$D^{\mathrm{cm}} - \sum_{\mathrm{k}} y_{\mathrm{gc,k}} \mathrm{AV^{cm}_{gc,k}} = 0. \qquad (\kappa) \tag{6.6}$$

The marginal social welfare impact capacity demand curve is constructed by evaluating Problem (6) for different amounts of capacity, D^{cm}: $\kappa(D^{\mathrm{cm}}) = \mathrm{argmax(Problem\ (6))}$.

Moving from an idealized system to a system with a deficit in capacity, e.g. because investors act risk-averse, extra payments to promote capacity are required. To reflect the idealized value of capacity, we propose to reflect these payments with the social value of capacity (κ). This way, the consumers

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would pay the marginal social welfare benefit of this capacity as a capacity price. In other words, the marginal costs for installing an additional MW of capacity would be equal to the marginal benefit. Vice-versa, this CDC would not distort the equilibrium if the available capacity equals the socially optimal level of capacity (i.e., when the reliability standard is met) as the marginal value of additional capacity would be 0. In the following sections, we compare this hypothetical curve against established capacity demand curve definitions.

IV. Stylized comparison

Assuming idealized conditions during a capacity shortage, wholesale electricity prices would rise to the VOLL [2]. However, in practice, price caps can hinder the price to reach the consumers' willingness to pay for electricity leading to involuntarily load shedding. In this section, we aim to qualitatively reproduce such situations by considering a price cap that is set to 3000 $\frac{€}{\mathrm{MWh}}$. With the VOLL being above the price cap, a capacity market is necessary to ensure meeting reliability standards by extra payments even when investors act risk-neutral. If investors act risk-averse, the need for extra payments is exacerbated as extra revenues are required to cover their risk premia.

For cross-comparing the systems that materialize under different CDCs, using the Social Welfare (SW) as a metric is not straightforward, as it considers the integrated WTP-function of the different capacity demand curves. For that reason, we compare the expected total system costs considering costs for investments and operation as well as costs for curtailing energy demand:

$$\mathbb{E}(\mathrm{SC}) = \sum_{\mathrm{k}} y_{\mathrm{gc,k}}\mathrm{CAPEX_k} + \sum_{\mathrm{k,t,s}} \mathrm{W_t P_s e_{gc,k,t,s} OPEX_{k,t,s}} + \underbrace{\sum_{\mathrm{t,s}} \mathrm{W_t P_s}(\overline{D^{\mathrm{e}}_{\mathrm{t,s}}} - \mathrm{e_{co,t,s}})\mathrm{VOLL}.}_{\text{cost of unserved energy (CUE)}} \quad (7)$$

In Section IV-A, we first establish the shapes of the demand curves assuming various degrees of risk aversions and estimations of the VOLL. Subsequently, these capacity demand curves are used as an input to perform a comparison in terms of capacity and system costs, firstly assuming perfect information (Section IV-B), and secondly considering estimation errors on VOLL and risk aversion (Section IV-C).

To keep the calculations lean and computationally tractable, we deliberately solely consider uncertainty in the electricity demand focusing on inherent biases in the definition of considered CDCs. Three different electricity demand scenarios (95%, 100%, 105%) are used based on the residual electricity demand of Belgium in 2019, implicitly accounting for renewable energy. The scenarios are created by scaling the reference scenario. In each scenario, 8760 time steps are considered with a duration of one hour. On the production side, two technologies are used, a base-load technology and a peak-load technology (Table I). All availability and capacity factors are assumed to be one. The risk aversion weight is assumed to be $\gamma = 0.5$.

Table I: Technology parameters

	base	peak
CAPEX[1] [€/MW]	128000	92000
OPEX [€/MWh]	62	90

[1] annualized

A. Long-run capacity demand curves

To compute long-run equilibria we apply Algorithm 1 to various system settings with varying VOLL (VOLL $\in$ [3000 €/MWh, 25000 €/MWh]) and the risk aversion parameter ($\beta \in [0.2, 1.0]$). The corresponding CDCs are depicted in Fig. 3. Here, the shape of each curve is illustrated for each (VOLL, β)-pair. Furthermore, market-clearing prices and quantities are depicted (by crosses). Possible price-quantity ranges introduced by risk aversion are grouped for each VOLL (shaded areas). The resulting curves allow observations connected to the impact of the VOLL, risk aversion, and coupled effects between both. Therefore, the impact of increasing VOLL shall be discussed first disregarding the effect of increasing risk aversion and vice-versa, before coupled effects are explained.

Increased VOLLs lead to higher target volumes for all CDCs (compare, e.g. the solid curves in Fig. 3). For the inelastic capacity demand, the capacity target increases towards the peak demand with decreasing step size if the VOLL increases. Equation (1) directly connects VOLL with LOLE, indicating an asymptotically decrease for increasing VOLLs. By definition, the MSWI curves' x-axis intersections coincide with the reliability standard, which shifts to the right for increasing VOLLs, while the slopes follow the marginal change in SW. Contrasting, the x-axis intersections of the MRI curve remain identical with increasing VOLL. However, slopes increase with the VOLL. For the SMRI curve, the net-CONE coincides with the capacity price, which increases with VOLL given the increasing investments needed. The elastic CDC is scaled by the net-CONE on the one hand and shifted to higher levels of capacity being defined by the RS as a function of the VOLL on the other hand. The semi-elastic CDC's x-axis intersections coincide with those of the inelastic demand curves equalling the RS of respective VOLLs. As the SMRI curves, the semi-elastic curves' breakpoint (B_3) increases with net-CONE that increases with higher levels of VOLL.

While all considered curves are defined taking the VOLL directly or indirectly into account, risk aversion does not impact all considered capacity curve definitions. The inelastic CDC, the MSWI, and the MRI CDCs are not defined as taking the net-CONE into the account and, therefore, their definitions are not impacted by risk aversion. However, market outcomes are dependent on the degree of risk aversion as market-clearing prices rise with increasing risk aversion (more risk-averse outcomes are illustrated with smaller markers). Additionally, for the MSWI and the MRI curves the price elasticity of the capacity demand leads to less installed capacity as risk-averse investors require higher capacity payments for the same installed capacity. The SMRI, elastic, and semi-elastic CDCs are defined taking the net-CONE into account. Therefore, increasing risk aversion leads to a shift of these curves to higher

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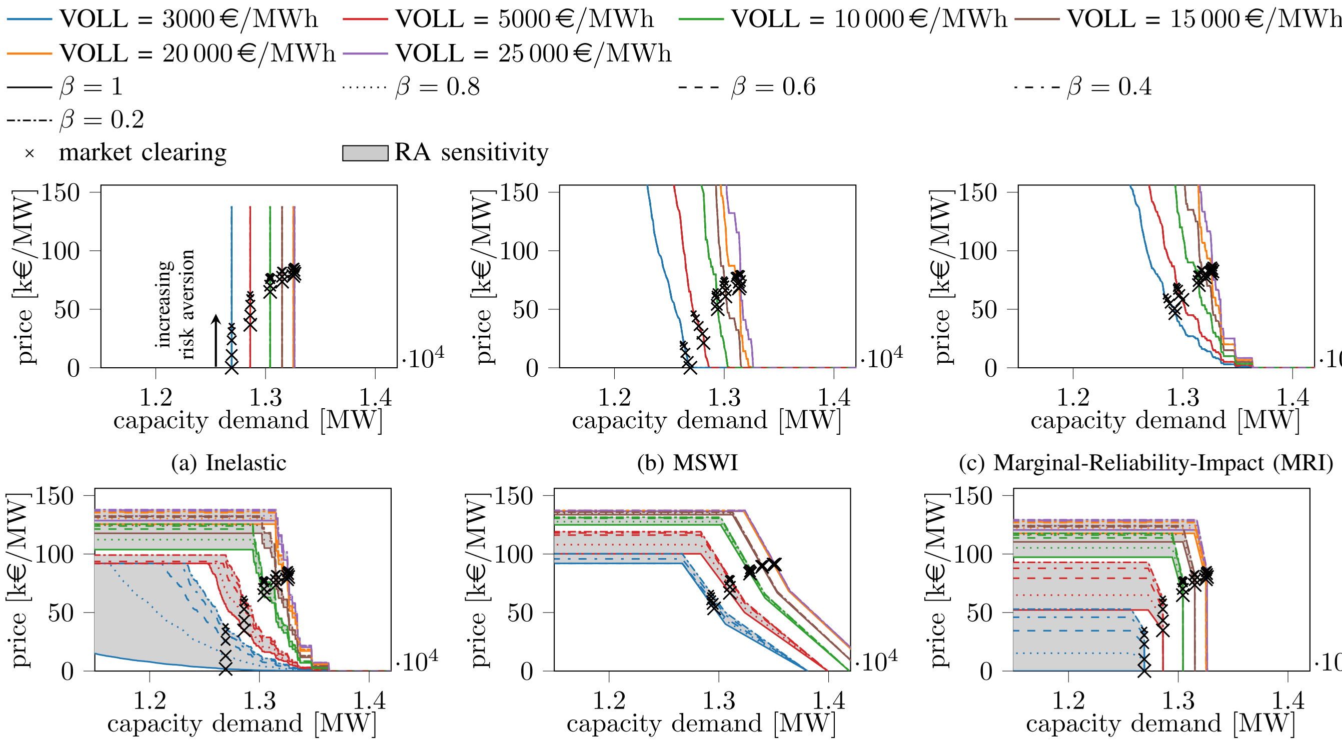

(a) Inelastic (b) MSWI (c) Marginal-Reliability-Impact (MRI)

(d) Scaled MRI (SMRI) (e) Elastic (f) Semi-elastic

Figure 3: CDCs in the long-term equilibrium considering varying VOLL and risk aversion. Shaded areas indicate feasible price-quantity spaces caused by risk aversion for a particular VOLL, indicated by the color of the curves. Market equilibria for different degrees of risk aversion are indicated with varying marker sizes, where the marker size decreases with increasing risk aversion.

prices. Strikingly, the long-term equilibria of the semi-elastic and SMRI CDCs do not show a dependency on risk aversion in terms of procured capacity. By applying Algorithm 1 to align the long-run equilibrium with the CDC definition, the capacity market always clears at (reliability target, net-CONE). Therefore, these curves can be understood as self-adjusting to risk aversion. The elastic CDC is also self-adjusting, however, the net-CONE is implicitly defined between P_2 and P_3, which is not the reliability target. Hence, the amount of procured capacity can deviate from the RS.

Furthermore, the impact of risk aversion is more pronounced for low VOLL values, both in terms of cleared capacity (MSWI, MRI, elastic) and capacity prices (all curves). With higher VOLLs, the target capacity moves closer to the peak electricity demand. Therefore, additional units increasingly rely on the capacity market to recover their investment costs. Risk aversion has, in our model set-up, a smaller impact on market outcomes with high VOLLs in terms of cleared capacity and capacity prices. Consequently, price-quantity spaces (grey shaded areas) shrink for higher VOLLs. The self-adapting curves (SMRI, semi-elastic) show the same price spread as the inelastic capacity demand curve as these, by definition, coincide with the inelastic market outcomes in terms of capacity price as well as volume. Compared to these market outcomes, results show that risk aversion impacts the MRI and elastic capacity demand curves to a lesser degree in terms of market prices and volumes.

In what follows, the total system cost associated with the CDCs presented in Fig. 3 are first evaluated assuming a perfect information bases (Section IV-B). As correct estimation of the VOLL and the degree of risk-aversion poses a notoriously challenging task, secondly an evaluation is performed assuming estimation errors (Section IV-C).

B. Performance with perfect information

Table II presents the system optimal outcomes without considering estimation errors on the degree of risk-aversion and the VOLL. The capacity market clearing points can also be found in Fig. 3. In principle, every capacity demand curve can procure optimal levels of capacity at the needed level of capacity remuneration if tuned correctly. However, here an inelastic capacity curve is used to provide optimal system outcomes that serve as a reference. The inelastic CDC is tuned according to the reliability standard as presented in Eq. (1). The increase in total capacity follows an asymptotic curve reaching the peak electricity demand with a VOLL that approaches infinity. Additional capacity is promoted by the capacity market (if the VOLL is above the price cap). Therefore, in this study, only the capacity of the peak-load technology increases with increasing VOLL. Connected to the installed total capacity and the shape of the load duration curve, EENS and LOLE decrease with increasing installed capacity. The system costs rise with higher VOLLs, being impacted by higher costs for capacity on the one hand and

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Table II: System optimal results. Outcomes are calculated using an inelastic capacity demand curve for varying risk aversion and VOLL with an energy price cap of $\overline{\lambda^{\mathrm{e}}} = 3000\ \frac{€}{\mathrm{MWh}}$.

	VOLL / β	$3\ \frac{\mathrm{k€}}{\mathrm{MWh}}$	$5\ \frac{\mathrm{k€}}{\mathrm{MWh}}$	$10\ \frac{\mathrm{k€}}{\mathrm{MWh}}$	$15\ \frac{\mathrm{k€}}{\mathrm{MWh}}$	$20\ \frac{\mathrm{k€}}{\mathrm{MWh}}$	$25\ \frac{\mathrm{k€}}{\mathrm{MWh}}$
total capacity [MW]	1.0	12692	12861	13042	13151	13253	13264
EENS [MWh]	1.0	8776.0	4479.0	1920.0	988.6	458.7	415.8
LOLE [h]	1.0	31.3	18.7	9.0	6.0	4.33	3.67
System costs [1e9 €]	1.0	6.14039	6.15241	6.16613	6.17181	6.17563	6.17782
	0.8	6.1404	6.15243	6.16615	6.17183	6.17565	6.17784
	0.6	6.14047	6.1525	6.16622	6.1719	6.17572	6.1779
	0.4	6.14059	6.15261	6.16633	6.17202	6.17584	6.17802
	0.2	6.14067	6.15269	6.16641	6.1721	6.17592	6.1781

decreasing EENS, on the other hand. However, increased investment and operational costs dominate over savings on EENS at system optimum. Risk aversion does not impact the amount of total installed capacity as the inelastic capacity demand enforces the social optimal capacity. However, risk aversion leads to less investment in base-load capacity, which is replaced by peak-load capacity [15]. This shift to more peak-load capacity is visible in terms of system costs.

Figure 4 compares considered CDCs with the socially optimal outcome by showing the differences to system-optimal outcomes (Table II). By definition, the SMRI, the semi-elastic, and the inelastic CDC procure the same amounts of capacity. The ideal performance of the SMRI and the semi-elastic curve, in this case, is grounded on the perfect estimation of the net-CONE. The elastic and the MRI CDC overprocure capacity given that the demand curves assign higher prices to capacity (Fig. 3). However, due to the capacity target moving towards the peak demand with increasing VOLL, the MRI CDC leads to outcomes that coincide with the social optimal capacity for high VOLLs. Contrasting, the MSWI approach underprocures capacity, particularly if the VOLL is evaluated above the electricity price cap. Due to the definition, the MSWI works best if the price cap is evaluated at VOLL. For VOLLs above the price cap, MSWI curve is constructed by externalizing the marginal costs of capacity assuming this specific VOLL, ignoring the price cap. Therefore, extra payments needed to procure capacity in case of a price cap below the VOLL appear to be insufficient to guarantee the reliability standard. Generally, risk aversion impacts only those approaches that do not adapt for the net-CONE at the reliability standard. For the other approaches, the results show that the impact of risk aversion becomes smaller with high values of VOLL (cf. Section IV-A).

In line with trends shown for installed capacities, total system costs of the SMRI and the semi-elastic approach are optimal from a system perspective. The MRI approach yields higher systems costs with low VOLLs as capacity is overprocured. However, as installed capacities converge towards the optimal capacities for higher VOLL values, system costs follow. The MSWI approach shows increased system costs by underprocuring capacity and the elastic approach by overprocuring capacity. The increase in deviation of the elastic CDC to the cost-optimum can be explained by the notion of increased costs for investments and operation versus saved curtailment costs (cf. Fig. 1).

C. Performance with imperfect information

To test the performance of the curves considering imperfect parametrization, in this section, expectations imposed in the planning phase of the CDCs deviate from the realizations. For each combination of VOLL and degree of risk aversion that was used in the planning phase (i.e. the definition of the CDC), the risk-averse generation expansion planning problem (Problem 2) is solved for all possible VOLLs (VOLL $\in [3000\ €/\mathrm{MWh}, 25000\ €/\mathrm{MWh}]$) and degrees of risk aversion ($\beta \in [1.0, 0.2]$) leading to 5400 solved problem instances, each considering uncertainty on the electricity demand (900 different problem instances per capacity demand curve). The capacity demand curves, used for the study, are created according to the VOLL and risk aversion expected in the planning phase (i.e., Fig. 3), not the realization.

Figure 5 summarizes the performance of the considered CDCs in terms of robustness against erroneous assumptions in the planning phase. The first quadrant shows data for all VOLL expectations as differences to the optimal system outcome to isolate the impact on system costs and optimal capacity. To give an example: a case with an expected VOLL $= 5000\,\frac{€}{\mathrm{MWh}}$ and a risk aversion of $\beta = 0.5$ that was evaluated with VOLL $= 10\,000\,\frac{€}{\mathrm{MWh}}$ and $\beta = 1.0$ is compared to a case where planning and realization coincide with VOLL $= 10\,000\,\frac{€}{\mathrm{MWh}}$ and $\beta = 1.0$ for the inelastic capacity demand curve. The reference case mimics a case assuming perfect foresight and, thus, serves as an optimal benchmark (as in Section IV-B). Note that absolute system costs (without referring to the optimal case) would form six distinct curves, one curve for each considered VOLL, similar to Fig. 1. The second and fourth quadrant illustrate ranges for systems costs and VOLLs depending on the VOLL assumed while planning the CDC. Displayed ranges indicate minimal and maximal values. The results from Fig. 4 are repeated for risk-neutral (RN) investors with perfect foresight (p.f.) to compare the impact of erroneous parametrization of the CDC with the idealized setting.

The results in Fig. 5 allow two common conclusions across all CDCs. First, an improper parameterization of the CDC, particularly a wrong estimation of the VOLL, yield higher system costs differences than those associated with the choice of CDC (Fig. 4), highlighting the need for both the analysis

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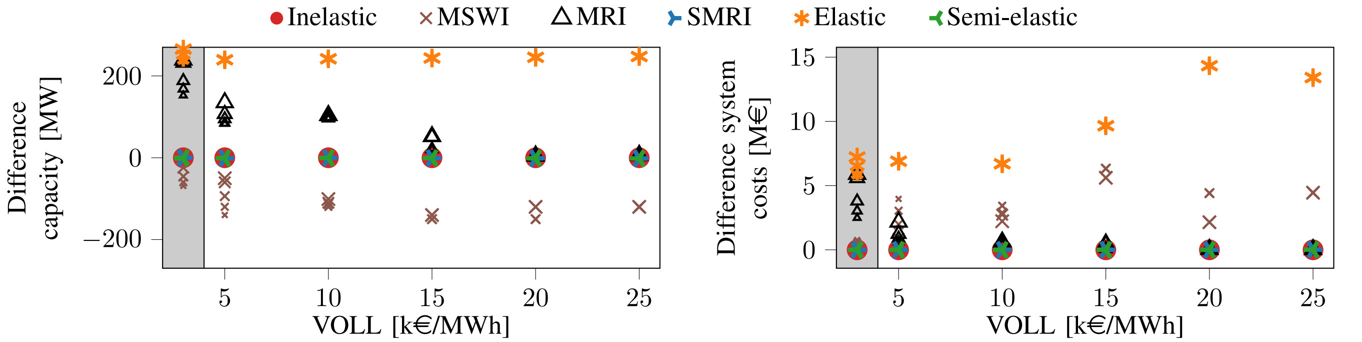

(a) Installed capacities (b) System costs

Figure 4: Differences of long-term installed capacity and system costs to system-optimal outcomes (Table II) with perfect foresight considering an energy price cap of $\overline{\lambda^{\mathrm{e}}} = 3000\ \frac{€}{\mathrm{MWh}}$. Risk aversion and VOLL are varied. Increasing levels of risk-averse outcomes are indicated with decreasing marker sizes. Shaded areas highlight the case where the price cap equals the VOLL.

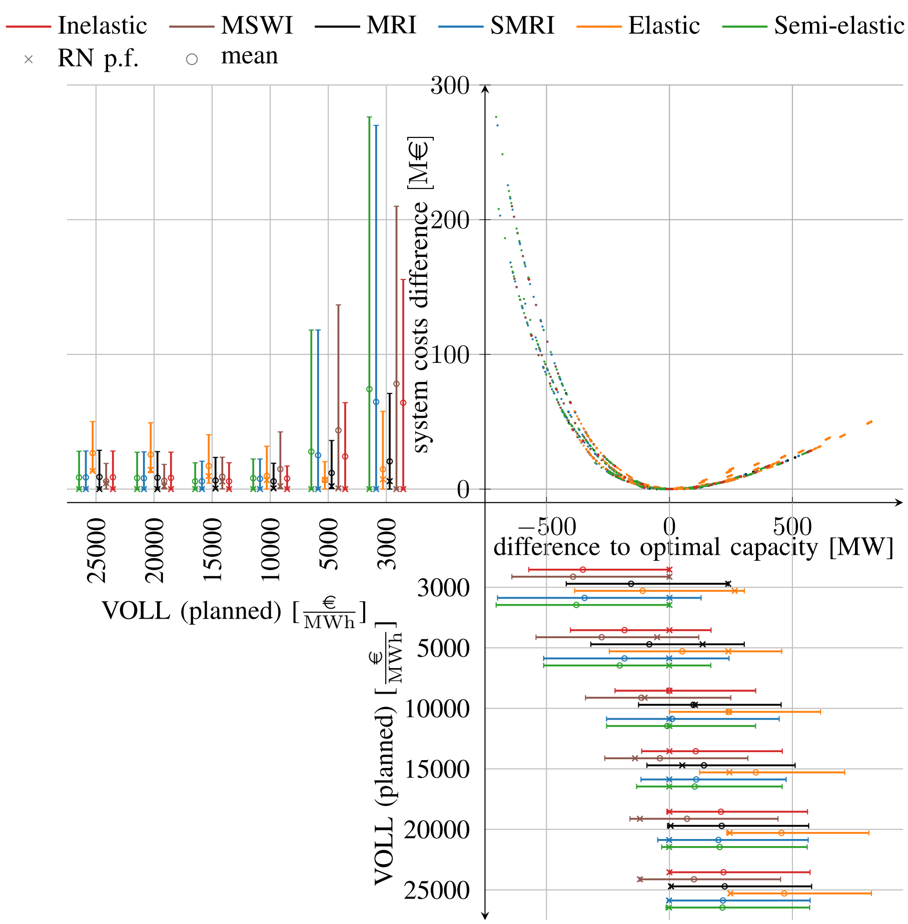

Figure 5: Performance comparison considering uncertain realizations. VOLL and RA used for the realization are varied from those used during the planning phase. All data points are plotted with respect to optimal outcomes with perfect foresight. The error bars show ranges for system costs and installed capacity indicating the mean value (circle) and a comparison to the risk-neutral outcome with perfect foresight (RN p.f., cross)

under perfect and imperfect information. As expected, our analysis confirms that underestimating the true VOLL can lead to higher costs triggered by underinvestments and high curtailment costs. Conversely, assuming too high VOLLs can lead to overinvestments, hence, excessive investment costs. However, as indicated in the second quadrant, underinvestments lead to higher system costs than overinvestments. Second, for low VOLLs, the uncertainty on the installed capacity scales with the elasticity of the capacity demand curve. While the inelastic CDC (red) has the smallest capacity range, the ranges of price elastic CDCs (semi-elastic (green), SMRI (blue)) are higher. However, these differences fade out with high VOLLs as risk aversion has a smaller impact with high VOLLs, because a larger share of revenues is secured via the (certain) capacity market.

Additionally, capacity demand curve-specific conclusions can be drawn. By procuring a fixed amount of generation capacity, the inelastic CDC has the smallest capacity and system cost range compared to those curves that do not have a bias to overprocurement of capacity (semi-elastic, SMRI curves). The MSWI approach tends to underprocure capacity with perfect foresight and high VOLLs (cf. Fig. 4). Due to this fact, it shows the lowest cost variation at high VOLLs, causing lower mean costs at high VOLLs and higher system costs at low VOLLs compared to the other approaches. In contrast, the MRI and the elastic approach tends to overprocure generation capacity in our analysis. However, as shown in Section IV-B, this tendency of the MRI approach to overprocure generation capacity fades out with high VOLLs. Therefore, this approach combines protection against underprocurement at low VOLLs in the CDC definition (even if high VOLLs are realized) while procuring system optimal levels of capacity at high VOLLs, resulting in cost-robust outcomes across the studied range of VOLLs and levels of risk aversion. In contrast, our results indicate that the elastic CDC may still lead to a systematic overprocurement of capacity at high VOLLs, which may cause elevated costs. Lastly, the semi-elastic and SMRI curves yield comparable results. While clearing at social optimal capacity with perfect foresight, our results indicate that both types of curves are the most sensitive to uncertainties (of the considered CDCs) when expecting low VOLLs (cf. VOLL (planned) = 3000 €/MWh). Comparing the inelastic CDC, for these curves, the impact of the expected degree of risk aversion and, therefore, the expected value of the net-CONE becomes eminent in terms of system costs.

V. Conclusion

The presented work provides a coherent comparison of selected capacity demand curves (CDCs), namely, an inelastic CDC, a Marginal-Reliability-Impact (MRI) curve as proposed by Zhao et al. [25], and implementations based on the principles of real-life CDCs of ISO-NE, PJM, and Belgium (referred to as SMRI, elastic and semi-elastic curves). The comparison is conducted assuming perfect as well as imperfect information bases considering risk-averse investors. Additionally, a hypothetical capacity demand curve (MSWI) based on the marginal social welfare gain of additional capacity is proposed and compared against established definitions. To not bias the demand curve definitions by system-dependent net costs of new entry (net-CONE), an iterative approach is proposed describing the system-dependent net-CONE and CDCs in the long-term equilibrium.

In a stylized case study, we show that effects stemming from risk aversion decrease at high VOLL levels. Assuming perfect foresight, we show that sloped capacity demand curves, defined with the net-CONE at the capacity that ensures the reliability standard is met (semi-elastic and SMRI curves), perform ideally in the long run. However, these approaches are sensitive to imperfect parameterization leading to higher system costs if estimation errors occur. These effects are most pronounced if the CDC is parameterized with an underestimated VOLL (erroneous foresight of the VOLL). The hypothetical capacity demand curve performs well if the VOLL is equal to the price cap, while payments needed to ensure the reliability standard appear to be insufficient if the VOLL is evaluated above the price cap. The MRI approach shows increased cost stability at marginally higher investment costs (perfect foresight), particularly at low expected VOLLs, while performing ideally with high VOLLs. Assuming estimation errors, we find that the MRI approach is robust to erroneous foresight. Lastly, the elastic capacity demand curve shows systematic overprocurement in our stylized case study.

To isolate the effects originating from CDC designs when considering risk aversion, this contribution assumes price-taking behavior and implicitly accounts for renewable energy. Future work could consider strategically motivated bids to assess the resilience of different CDC designs against possible market power abuse. Additionally, accounting for the effect of increased demand elasticity, e.g., introduced by short-term storage, and uncertain and system-dependent capacity or derating factors of short-term storage and renewable technologies pose an interesting extension of this work.

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